

Numerical BILP Approximation Summary

			Maximization Summary													
			R Relaxed LP Approximation						R Collapsed Dual Approximation				Python Collapsed Dual Approximation			
			Exact	Beasley's	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution
			Maximum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
Small Textbook Maximization Problem	Variables 4	Constraints 3	19	NA	19	22	0.8636	0.0015	19	22	0.8636	0.0475	19	22	0.8636	0.0114
			Mirror-Image Minimization Summary													
			R Relaxed LP Approximation						R Collapsed Dual Approximation				Python Collapsed Dual Approximation			
			Exact	Beasley's	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution
			Minimum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
			23	NA	23	21.7250	1.0587	0.0015	23	21.7249	1.0587	0.0029	23	21.7231	1.0588	0.0100
			Maximization Summary													
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			Exact	Beasley's	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution
			Maximum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
Beasley Problem mknap1_7	Variables 50	Constraints 5	16537	NA	16436	16612.82	0.9894	0.0034	16436	16612.83	0.9894	0.0041	16436	16613.95	0.9893	0.0040
			Mirror-Image Minimization Summary													
			R Relaxed LP Approximation						R Collapsed Dual Approximation				Python Collapsed Dual Approximation			
			Exact	Beasley's	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution
			Minimum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
			11162	NA	11384	10926.99	1.0418	0.0037	11384	10926.99	1.0418	0.0037	11384	10907.52	1.0437	0.0040
			Maximization Summary													
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			Exact	Beasley's	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution
			Maximum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
Beasley Problem mknapcb9_30	Variables 500	Constraints 30	NA	300460	299930	301020.6	0.9964	0.0690	300016	301020.6	0.9967	0.1017	299649	301021.9	0.9954	0.0199
			Mirror-Image Minimization Summary													
			R Relaxed LP Approximation						R Collapsed Dual Approximation				Python Collapsed Dual Approximation			
			Exact	Beasley's	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution
			Minimum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
			NA	NA	259089	257611.5	1.0057	0.0587	259079	257611.4	1.0057	0.0844	259970	257609.7	1.0092	0.0409
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			Exact	Beasley's	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution
			Maximum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
Large Randomized Problem	Variables 50000	Constraints 300	NA	NA	NA	NA	NA	NA	47193	47240.26	0.9990	3433.5	47193	47240.26	0.9990	392.2
			Mirror-Image Minimization Summary													
			R Relaxed LP Approximation						R Collapsed Dual Approximation				Python Collapsed Dual Approximation			
			Exact	Beasley's	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution
			Minimum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
			NA	NA	NA	NA	NA	NA	2363738	2363669	1.0000	3452.5	2363740	2363689	1.0000	391.8

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				Exact	Beasley's	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution
				Minimum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
Small Textbook Minimization Problem		Variables	Constraints	20	NA	22	18	1.2222	0.0026	22	18	1.2222	0.0040
		9	5										
						Mirror-Image Maximization Summary							
						R Relaxed LP Approximation					R Collapsed Dual Approximation		
				Exact	Beasley's	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution
				Maximum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
				23	NA	23	23	1.0000	0.0026	23	23	1.0000	0.0021
						Minimization Summary							
						R Relaxed LP Approximation					R Collapsed Dual Approximation		
				Exact	Beasley's	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution
				Minimum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
Beasley Problem scp5		Variables	Constraints	5	5	6	3.3908	1.7695	0.0609	7	3.3831	2.0691	0.3075
		500	50										
						Mirror-Image Maximization Summary							
						R Relaxed LP Approximation					R Collapsed Dual Approximation		
				Exact	Beasley's	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution
				Maximum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
				6	NA	6	8.3659	0.7172	0.0458	6	8.3680	0.7170	0.5981
						Minimization Summary							
						R Relaxed LP Approximation					R Collapsed Dual Approximation		
				Exact	Beasley's	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution
				Minimum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
Beasley Problem scp5		Variables	Constraints	236	240	246	234.8889	1.0473	3.570	263	226.9616	1.1588	8.459
		3000	300										
						Mirror-Image Maximization Summary							
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				Exact	Beasley's	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution
				Maximum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
				NA	NA	7767	8313.7	0.9342	3.418	7620	8386.3	0.9086	31.793
						Minimization Summary							
						R Relaxed LP Approximation					R Collapsed Dual Approximation		
				Exact	Beasley's	BILP	Lower	Approx	Execution	BILP	Lower	Approx	Execution
				Minimum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
Beasley Problem scpnr5		Variables	Constraints	NA	28	31	21.3219	1.4539	26.853	30	14.6637	2.0459	38.005
		5000	500										
						Mirror-Image Maximization Summary							
						R Relaxed LP Approximation					R Collapsed Dual Approximation		
				Exact	Beasley's	BILP	Upper	Approx	Execution	BILP	Upper	Approx	Execution
				Maximum	Best	Approx	Bound	Ratio	Seconds	Approx	Bound	Ratio	Seconds
				NA	NA	238	1021.0	0.2331	28.118	226	1045.7	0.2161	268.125
J. E. Beasley's problems were downloaded from his OR-Library at people.brunel.ac.uk/~mastjjb/jeb/info.html .													
Logical, precise results for multidimensional knapsack problems and their mirror images.													
Less so for set covering problems and their mirror images.													
Approximation ratios are relatively useless for Beasley's set covering problems.													
All code was executed on a Dell XPS Tower PC.													